FACULTY
OF MATHEMATICS
AND PHYSICS
Charles University

MASTER THESIS

Bc. Kamil Belán

Numerical methods for the simulation of air flow past a mountain

Mathematical Institute

Supervisor of the Thesis doc. RNDr. Michal Pavelka, Ph.D.
Study Programme Mathematical and Computational Modelling in Physics

Prague 2025
(Draft - March 11, 2026)

Declaration of Authorship

I declare that I carried out this master thesis on my own, and only with the cited sources, literature and other professional sources. I understand that my work relates to the rights and obligations under the Act No. 121/2000 Sb., the Copyright Act, as amended, in particular the fact that the Charles University has the right to conclude a license agreement on the use of this work as a school work pursuant to Section 60 subsection 1 of the Copyright Act.

Place & Date *Bc. Kamil Belán*

Write dedication here.

Bc. Kamil Belán

Information

Title	Numerical methods for the simulation of air flow past a mountain
Author	Bc. Kamil Belán
Institute	Mathematical Institute
Supervisor	doc. RNDr. Michal Pavelka, Ph.D. — Mathematical Institute
Keywords	Manual, Demo, Draft, WIP

Abstract

Write abstract here.

Contents

Information & Abstract iv

Contents v

Introduction vi

1 Flows around a topography 1
- 1.1 Physical description . . . 1
  - Observed phenomeona 1 • Euler equations 1 • Thermodynamics 1 • Hydrostatic balance 1
- 1.2 Buoyancy oscillations . . . 1
  - Stratification 1 • Potential temperature 1
- 1.3 Mathematical modelling of atmospheric flows . . . 1
  - Initial and boundary conditions 1 • Discretization 1 • Well-balancing 2

2 SPH 3
- 2.1 Overview of SPH . . . 3
  - Pros and cons 3 • Uses 3
- 2.2 Cornerstones of SPH . . . 3
  - Partition 3 • Kernel interpolation 3 • Differential operators 3
- 2.3 SPH formulations . . . 3
  - WCSPH 3 • Hopkins 3 • Comparison of the formulations 5
- 2.4 Additional techniques . . . 5
  - Packing algorithm 5 • Adaptive smoothing length 5 • Initial configuration 5

3 Trixi 6
- 3.1 Finite Volume Method . . . 6

4 Voronoi 7

Summary and Outlook 8

References 9

I Introduction

1 Flows around a topography

1.1 Physical description

Observed phenomeona

Euler equations

$$\mathbb{T} = -p\mathbb{I} + 2\mu\mathbb{D},$$

where $p = p(\rho, \theta)$ is the thermodynamic pressure, $\mathbb{I}$ is the identity matrix, μ is the dynamic viscosity and $\mathbb{D} = \operatorname{sym} \boldsymbol{\nabla}\mathbf{v}$ is the symmetric part of the velocity gradient.

$$\partial_t(\rho\mathbf{v}) + \boldsymbol{\nabla} \cdot (\rho\mathbf{v} \otimes \mathbf{v}) = \boldsymbol{\nabla} \cdot \mathbb{T} + \rho\mathbf{b},$$

with $\mathbf{b}$ being the specific density of the body forces, *e.g.* gravity.

$$\partial_t\rho + \boldsymbol{\nabla} \cdot (\rho\mathbf{v}) = 0.$$

Thermodynamics

We have two equations of state:

$$p(\rho, \theta) = R\rho\theta,$$

$$e(\rho, \theta) = zR\theta.$$

Hydrostatic balance

1.2 Buoyancy oscillations

Stratification

Potential temperature

1.3 Mathematical modelling of atmospheric flows

Initial and boundary conditions

Discretization

- SPH

- FVM
- Voroni
- Discontinuous Galerkin

Well-balancing

2 SPH

2.1 Overview of SPH

Pros and cons

Uses

2.2 Cornerstones of SPH

Partition

Kernel interpolation

Differential operators

2.3 SPH formulations

WCSPH

Hopkins

Pressure - entropy

Recall that the entropy of an isentropic flow of an ideal gas has the form

$$\eta = \eta_0 + c_v \log\left(\frac{P}{\rho^\gamma}\right)$$

and $\eta =$ const. We see that the quantity A defined as [1]

$$A = \frac{P}{\rho^\gamma},$$

is in fact equal to

$$A = \exp\left(\frac{\eta - \eta_0}{c_v}\right),$$

meaning that the flow is *isentropic*, if and only if the quantity A is *conserved* under the flow. Thus A is an *entropy-like* variable: conserved under isentropic flows.

The choice

$$x_i = m_i A_i^{\frac{1}{\gamma}}, \tag{2.1}$$

gives the pressure P_i as

$$P_i = \left(\sum_{j=1}^{N} m_j A_j^{\frac{1}{\gamma}} w_{ij}(h_i) \right)^{\gamma} \tag{2.2}$$

and the evolution equations as

$$\frac{\mathrm{d}\mathbf{v}_i}{\mathrm{d}t} = -\sum_{j=1}^{N} m_j (A_i A_j)^{\frac{1}{\gamma}} \left(\frac{f_i P_i}{P_i^{\frac{2}{\gamma}}} \nabla_i w_{ij}(h_i) + \frac{f_j P_j}{P_j^{\frac{2}{\gamma}}} \nabla_i w_{ij}(h_j) \right), \tag{2.3}$$

with

$$f_i = \left(1 + \frac{h_i}{2n_i} \frac{\partial n_i}{\partial h_i} \right)^{-1}$$

Pressure - potential temperature

Recall the *potential temperature* θ

$$\theta = T \left(\frac{P_0}{P} \right)^{\frac{R}{c_p}},$$

where T is the gas's thermodynamic temperature, and P_0 is the reference pressure.[1] Using the thermal state equation

$$P = R\rho T,$$

together with the fact

$$\frac{R}{c_p} = \frac{\gamma - 1}{\gamma},$$

we obtain

$$\theta = \frac{P}{\rho R} P_0^{\frac{\gamma-1}{\gamma}} P^{\frac{1-\gamma}{\gamma}} = \frac{P_0^{\frac{\gamma-1}{\gamma}}}{R} \left(\frac{P^{\frac{1}{\gamma}}}{\rho} \right) = \frac{P_0^{\frac{\gamma-1}{\gamma}}}{R} \left(\frac{P}{\rho^{\gamma}} \right)^{\frac{1}{\gamma}} = \frac{P_0^{\frac{\gamma-1}{\gamma}}}{R} A^{\frac{1}{\gamma}},$$

[1] Potential temperature is the thermodynamic temperature the air parcel would possess if *adiabatically* moved from the thermodynamic state (T, P) to the state (θ, P_0).

so setting

$$\tilde{\theta}_0 := \frac{P_0^{\frac{\gamma-1}{\gamma}}}{R}$$

we obtain

$$\theta = \tilde{\theta}_0 A^{\frac{1}{\gamma}}. \tag{2.4}$$

This implies the potential temperature θ is also *an entropy-like* variable - it is *conserved* under isentropic flows.

Rewriting 2.1 using 2.4 we obtain

$$x_i = \frac{m_i}{\tilde{\theta}_0}\,\theta_i. \tag{2.5}$$

Using this quantity the pressure 2.2 can then be expressed as

$$P_i = \left(\sum_{j=1}^{N} \frac{m_j}{\tilde{\theta}_0}\,\theta_j w_{ij}(h_i)\right)^{\gamma}. \tag{2.6}$$

From 2.3 we obtain

$$\frac{\mathrm{d}\mathbf{v}_i}{\mathrm{d}t} = -\sum_{j=1}^{N} m_j\,\frac{\theta_i\theta_j}{\tilde{\theta}_0^2}\left(\frac{f_i P_i}{P_i^{\frac{2}{\gamma}}}\,\nabla_i w_{ij}(h_i) + \frac{f_j P_j}{P_j^{\frac{2}{\gamma}}}\,\nabla_i w_{ij}(h_j)\right), \tag{2.7}$$

with **??** unchanged

$$f_i = \left(1 + \frac{h_i}{2n_i}\,\frac{\partial n_i}{\partial h_i}\right)^{-1}. \tag{2.8}$$

Comparison of the formulations

2.4 Additional techniques

Packing algorithm

Adaptive smoothing length

Initial configuration

3 Trixi

3.1 Finite Volume Method

4 Voronoi

Summary and Outlook

Summary and Outlook.

References

Back-references to the pages where the publication was cited are given by •.

[1] Philip F. Hopkins. A General Class of Lagrangian Smoothed Particle Hydrodynamics Methods and Implications for Fluid Mixing Problems. *Monthly Notices of the Royal Astronomical Society*, **2013**.
DOI: 10.1093/mnras/sts210 arXiv: 1206.5006 [astro-ph]
(visited on 01/08/2026) 3